Banking Fraud Detection – Data Analysis Case Study

Business Scenario

A bank has observed an increase in suspicious transactions through online banking, ATMs, and POS terminals.

The bank's Risk & Fraud Analytics team has collected transaction-level data containing information about:

- Customer details
- Transaction amount
- Transaction type
- Account balance
- Device used
- Location changes
- Transaction time
- Failed transactions
- Fraud status

The bank wants to understand **what patterns are associated with fraudulent transactions** and identify suspicious behavior through data analysis.

You have been assigned as a **Junior Data Analyst** in the Fraud Analytics team.

Your task is to explore, clean, analyze, and visualize the transaction data and provide meaningful business insights.

Dataset: banking_fraud_data.csv

Each row represents one banking transaction.

Important Columns

Column	Description
Transaction_ID	Unique transaction identifier
Customer_ID	Unique customer identifier
Age	Customer age
City	Customer's city
Transaction_Amount	Amount involved in the transaction
Transaction_Type	Online, ATM, or POS transaction
Account_Balance	Customer's account balance
Device_Type	Device used for transaction
Location_Change	Whether the transaction occurred from a different location
Transaction_Time_Hour	Hour when transaction occurred
Is_Weekend	Whether transaction occurred on a weekend
Failed_Transactions_Last_24h	Number of failed transactions in previous 24 hours
Is_Fraud	Target variable: 1 = Fraud, 0 = Non-Fraud

Your Objective

The bank wants answers to the following questions:

"Can we identify transaction patterns that are strongly associated with fraud?"

You need to perform Exploratory Data Analysis (EDA) and provide evidence-based conclusions.

Your analysis should cover:

1. Data understanding
2. Data quality assessment
3. Data cleaning
4. Outlier detection
5. Statistical analysis
6. Data visualization
7. Fraud pattern analysis
8. Business insights
9. Recommendations

Phase 1 – Understand the Data

Load the dataset using Pandas.

Perform the following:

- Display the first 5 rows.
- Display the last 5 rows.
- Find the number of rows and columns.
- Display column names.
- Check data types.
- Generate dataset information using info().
- Generate statistical summary using describe().
- Identify categorical and numerical columns.
- Find the number of unique customers.
- Find the number of unique transactions.
- Find unique transaction types.
- Find unique device types.
- Find unique cities.

Questions

1. How many transactions are present?
2. How many customers are present?
3. What types of transactions are available?
4. Which columns are numerical?
5. Which columns are categorical?

Phase 2 – Data Quality Investigation

Before performing analysis, determine whether the data is reliable.

Check for:

Missing Values

Identify columns containing missing values.

Questions:

- Which columns contain missing values?
- How many missing values are present?
- What percentage of each column is missing?

Decide how you would handle the missing values.

Possible approaches:

- Mean
- Median
- Mode
- Forward fill
- Backward fill
- Removing records

Explain **why** you selected your approach.

Phase 3 – Identify Invalid Data

Look for values that do not make business sense.

For example:

Transaction Amount

Investigate:

- Negative transaction amounts
- Zero transaction amounts
- Extremely large transactions

Account Balance

Check whether:

- Account balance is negative
- Transaction amount is greater than account balance

Age

Check whether there are:

- Unrealistic ages
- Extremely low values
- Extremely high values

Transaction Time

Verify that:

Transaction_Time_Hour

is between:

0 and 23

Questions

1. Which records contain invalid values?
2. Should these records be removed or corrected?
3. What assumptions did you make?

Phase 4 – Outlier Detection

The bank is particularly interested in unusual transaction amounts.

Use visualization and statistical methods to identify outliers.

Method 1 – Boxplot

Create a boxplot for:

Transaction_Amount

Identify potential outliers.

Method 2 – IQR Method

Calculate:

- Q1
- Q3
- IQR
- Lower Bound
- Upper Bound

Formula:

IQR = Q3 - Q1

Lower Bound = Q1 - 1.5 × IQR

Upper Bound = Q3 + 1.5 × IQR

Identify transactions outside these limits.

Questions

1. How many outliers are present?
2. Which transactions are outliers?
3. Are all outliers fraudulent?
4. Should an outlier automatically be considered fraud?

Important Business Question

Is an outlier necessarily an error?

Think carefully before removing an extreme transaction.

A ₹10,00,000 transaction may be unusual, but it could still be a legitimate transaction.

Phase 5 – NumPy Analysis

Use NumPy to perform numerical analysis.

Calculate:

- Mean transaction amount
- Median transaction amount
- Standard deviation
- Minimum transaction amount
- Maximum transaction amount
- Percentiles

Compare:

Fraudulent Transactions

with

Non-Fraudulent Transactions

Calculate:

- Mean transaction amount
- Median transaction amount
- Standard deviation

Question

Is the average transaction amount higher for fraudulent transactions?

Support your answer with calculations.

Phase 6 – Fraud Distribution Analysis

Analyze the target variable:

Is_Fraud

Determine:

- Number of fraudulent transactions
- Number of legitimate transactions
- Fraud percentage

Create a visualization showing:

Fraud vs Non-Fraud

Question

Is the dataset balanced or imbalanced?

Why could class imbalance become a problem in a real fraud detection system?

Phase 7 – Fraud by Transaction Type

Analyze fraud across:

- Online
- ATM
- POS

Create an appropriate visualization.

Questions

1. Which transaction type has the highest number of fraud cases?
2. Which transaction type has the highest fraud rate?
3. Is online banking more risky than ATM/POS?

Phase 8 – Device Analysis

Analyze fraud based on:

- Mobile
- Laptop
- ATM
- Card

Create suitable visualizations.

Questions

1. Which device is associated with the highest fraud?
2. Does the device type appear to influence fraud?
3. What percentage of transactions from each device are fraudulent?

Phase 9 – Location Change Analysis

Analyze:

Location_Change

Compare:

- Location changed
- Location did not change

against:

Is_Fraud

Business Question

Are transactions involving a location change more likely to be fraudulent?

Create a visualization and calculate the fraud percentage for each group.

Phase 10 – Transaction Time Analysis

Analyze:

Transaction_Time_Hour

Create a new column:

Time_Category

Suggested categories:

Morning → 5–11

Afternoon → 12–16

Evening → 17–20

Night → 21–23

Late Night → 0–4

Analyze fraud across different time periods.

Questions

1. During which time period does fraud occur most frequently?
2. Are late-night transactions more suspicious?
3. Is transaction time a useful fraud indicator?

Phase 11 – Failed Transactions Analysis

Analyze:

Failed_Transactions_Last_24h

Compare failed transactions with fraud status.

Create a visualization.

Questions

1. Do fraudulent transactions have more failed attempts?
2. What is the average number of failed transactions for fraud cases?
3. Does a higher number of failed transactions indicate increased risk?

Phase 12 – Correlation Analysis

Select appropriate numerical columns and create a correlation matrix.

Use:

Seaborn Heatmap

Analyze relationships between:

- Transaction Amount
- Account Balance
- Age
- Transaction Time
- Failed Transactions
- Is_Fraud

Questions

1. Which variables have the strongest relationship with fraud?
2. Which variables appear unrelated?
3. Are there highly correlated independent variables?

Phase 13 – Visualization Dashboard

Create a mini analytical dashboard using Matplotlib and Seaborn.

Your dashboard should contain at least:

Visualization 1

Fraud vs Non-Fraud

Visualization 2

Fraud by Transaction Type

Visualization 3

Fraud by Device Type

Visualization 4

Transaction Amount Distribution

Visualization 5

Transaction Amount by Fraud Status

Visualization 6

Fraud by Time Category

Visualization 7

Fraud vs Location Change

Visualization 8

Correlation Heatmap

Phase 14 – Feature Engineering

Create meaningful new columns.

Feature 1 – Transaction Ratio

Transaction_Ratio =

Transaction_Amount / Account_Balance

This represents how large the transaction is compared with the customer's available balance.

Feature 2 – High Value Transaction

Create:

Is_High_Value

Example:

Transaction Amount > ₹1,00,000 → Yes

Otherwise → No

Feature 3 – High Risk Time

Create:

High_Risk_Time

For example:

0–4 → Yes

Otherwise → No

Feature 4 – High Failed Attempts

Create:

High_Failed_Attempts

For example:

Failed Transactions >= 3 → Yes

Otherwise → No

Phase 15 – Identify Suspicious Transactions

Create a simple rule-based analysis.

Flag a transaction as **Potentially Suspicious** if one or more conditions are satisfied:

- Very high transaction amount
- Location changed
- Transaction occurred during late night
- Multiple failed transactions
- Transaction amount is unusually high compared with account balance

Do NOT assume that these rules automatically mean fraud.

The objective is to identify transactions that deserve further investigation.

Final Business Questions

After completing the analysis, answer the following:

Question 1

What are the strongest indicators of fraudulent transactions?

Question 2

Are high-value transactions more likely to be fraudulent?

Question 3

Does a location change indicate higher fraud risk?

Question 4

Does transaction time influence fraud?

Question 5

Does the number of failed transactions correlate with fraud?

Question 6

Which transaction type has the highest fraud rate?

Question 7

Which device type has the highest fraud rate?

Question 8

Are outliers necessarily fraudulent?

Question 9

Which variables should the bank monitor more closely?

Question 10

What recommendations would you give to the bank?

Final Deliverable

Submit a Jupyter Notebook:

Banking_Fraud_EDA_<YourName>.ipynb

Your notebook should contain:

1. Introduction
2. Business Problem
3. Dataset Understanding
4. Data Quality Analysis
5. Data Cleaning
6. Outlier Detection
7. NumPy Analysis
8. Exploratory Data Analysis
9. Feature Engineering

10. Visualizations

11. Fraud Pattern Analysis

12. Key Insights

13. Business Recommendations

14. Conclusion

Expected Final Conclusion

At the end of the notebook, write a short business summary answering:

"Based on the data, what patterns should the bank monitor to identify potentially fraudulent transactions?"

Your conclusion should contain:

Key Findings

- Finding 1
- Finding 2
- Finding 3
- Finding 4
- Finding 5

Business Recommendations

- Recommendation 1
- Recommendation 2
- Recommendation 3